CMPG321 · ADVANCED DATABASES · PHASE 2



Where does the money go after a rider pays?

Driver earnings, platform commission and payout optimisation in South African ride-hailing

Database Design, Setup and Analysis

Group 36 · Data Alchemists

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Table of contents

The team **3**
1. Executive summary **3**
Why this matters 4
Objective and requirements 4
2. Database design, setup and loading **5**
Loading and validation 5
Setup procedure 6
Table design 6
Relationships 6
Tools used 7
3. SQL development and performance **8**
Design decisions in the SQL 8
Selected SQL 8
Performance testing (RQ4 queries) 9
4. Findings by research question **10**
RQ1: Fares, commission and payout by platform 10
RQ2: Commission and the share of the fare drivers keep 10
RQ3: Ride category 11
RQ4: Distance, duration, peak periods and surge pricing 13
RQ5: City and ride-category combination 15
RQ6: Net hourly earnings after estimated fuel 15
5. Actionable insights **18**
6. Limitations **18**
7. Conclusion **18**
Appendix A: Data dictionary **19**

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1. Executive summary

After a rider pays for an Uber or Bolt trip in South Africa, how much reaches the driver, and once fuel is counted, does it add up to a living wage? Six research questions follow the money from fare to wage. All analysis uses the 33,331 completed trips (of 50,000); the 16,669 cancelled trips carry zero distance, duration, and payout and are excluded.

```
THE AVERAGE TRIP, AS A RECEIPT

  Rider pays ............... R462.91
  - platform commission .... R99.38
  = DRIVER PAYOUT .......... R363.53
  - estimated fuel ......... R31.27
  ================================
  DRIVER KEEPS ............. R332.26

  33,331 completed trips. Before idle time and vehicle costs.
```

Headline findings:

- Bolt drivers keep about R26 more per trip than Uber drivers (R377.56 vs R351.25), mostly because Bolt takes less commission.
- Drivers keep 80.20% of the commissionable fare on Bolt (20% commission) and 75.26% on Uber (25%).
- Ride category barely matters: all four categories sit within R4.10 and about 79% of fare.
- Distance, duration and surge drive payout; tips do not. Peak trips pay more only because they carry more surge.
- City beats category: Cape Town takes the top four places, led by Cape Town Budget at R456.41 per trip.
- Estimated net pay is R262.52 per hour against the R30.23 benchmark, and every city clears it. Idle time and vehicle costs are not measured.

Why these matters

A South African Uber or Bolt driver hands 20-25% of every fare to the platform and still cannot tell whether they made real money once petrol is counted. A payout is not profit. We approach the data from an operational and financial perspective, following a completed trip from fare to commission to payout to fuel.

> ***Are the people driving South Africa's ride-hailing economy earning a living from it, or just staying afloat?***

Objective and requirements

Objective: measure how much of each fare reaches the driver, what is left after estimated fuel, and how this differs by platform, ride category, trip conditions and city, then test the result against South Africa's minimum-wage benchmarks.

2. Database design, setup and loading

The database is PostgreSQL (cmpg321_data_alchemists), restored from one backup file so every member works on an identical copy. It has seven linked tables, primary keys on every table, foreign keys for every relationship, 22 CHECK constraints on commission, ratings, trip status, ride category, and payment method, NOT NULL on key fields, NUMERIC types for rand amounts, TIMESTAMP for trip requests, and eight indexes on join, status, date, and category columns.

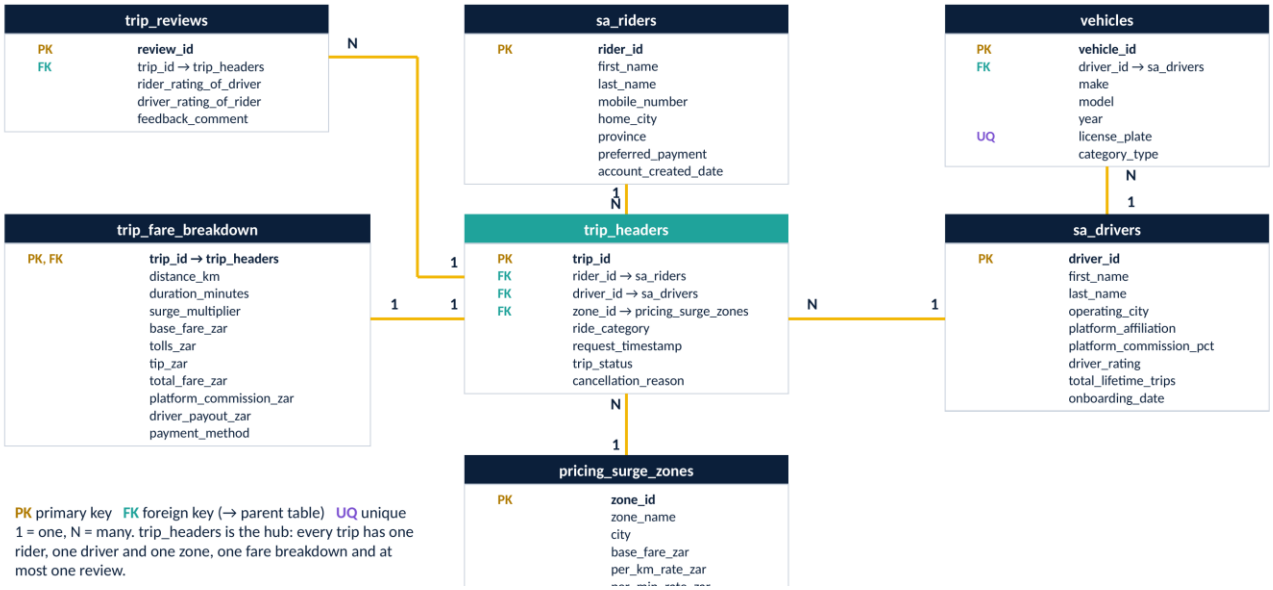


Figure 1: ERD with primary keys (PK), foreign keys (FK, with the parent table), the unique key (UQ), and cardinalities (1 = one, N = many).

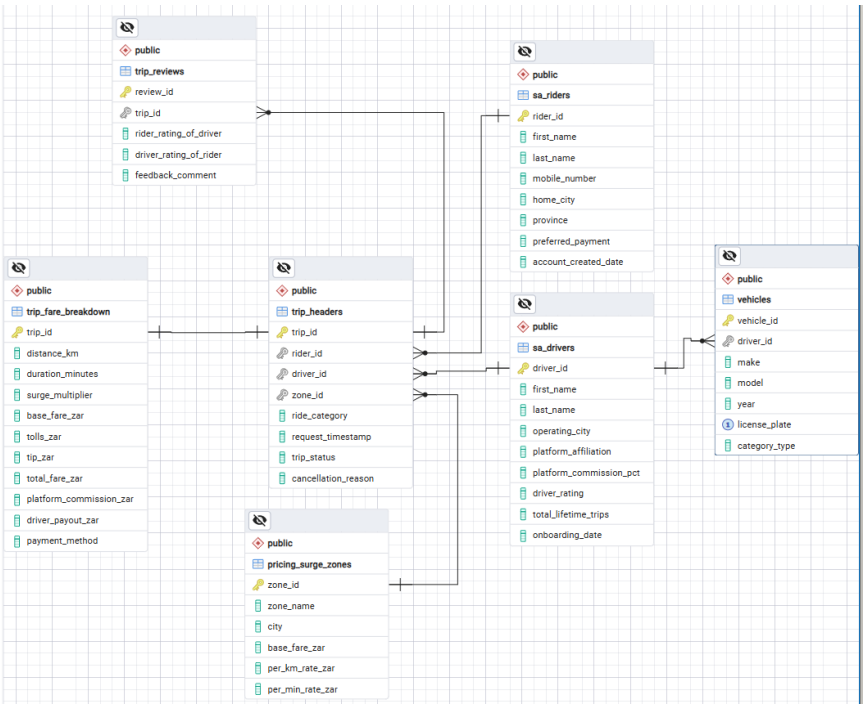


Figure 2: The same design as generated by pgAdmin 4 from the live PostgreSQL database.

Loading and validation

Table	Rows	Role
sa_drivers	800	Driver profile, platform and commission
vehicles	800	One vehicle per driver
sa_riders	1,200	Rider profile
pricing_surge_zones	9	Zone tariffs and city
trip_headers	50,000	One row per trip request
trip_fare_breakdown	50,000	Fare, commission and payout per trip
trip_reviews	25,055	Ratings (about half of trips)

The validation script (04_validation.sql) confirms these counts after every restore. The data covers 1 March 2025 to 31 March 2026 in seven cities. Of the 50,000 trips, 33,331 are completed and 16,669 are cancelled (8,417 by riders and 8,252 by drivers). On every completed trip, total fare minus commission equals driver payout.

Setup procedure

- 01_create_database.sql and 03_load_data.sql document how the master database is distributed; the group restores cmpg321_data_alchemists_initial.backup in pgAdmin 4 instead of importing CSV files by hand.
- 02_create_tables.sql creates the seven tables with primary keys, foreign keys, data types and constraints.
- 04_validation.sql counts the rows in every table and matches the source files exactly.
- 05_indexes.sql creates the eight indexes: trip_headers (driver_id, rider_id, zone_id, request_timestamp, trip_status, ride_category), trip_fare_breakdown (trip_id) and trip_reviews (trip_id).
- The original CSV files are kept in 07_Dataset; all analysis SQL is in 02_SQL_Queries, one file per research question plus additional_advanced_queries.sql.

Table design

Table	Primary key	Foreign keys	Main constraints
sa_drivers	driver_id	none	platform_affiliation limited to Uber / Bolt / Dual-Platform; commission 0-1; rating 0-5; lifetime trips >= 0
sa_riders	rider_id	none	Identity and payment profile; most descriptive fields optional
vehicles	vehicle_id	driver_id to sa_drivers	license_plate UNIQUE; year >= 1900
pricing_surge_zones	zone_id	none	Base fare and per-km and per-minute rates >= 0
trip_headers	trip_id	rider_id, driver_id, zone_id	ride_category limited to four categories; trip_status limited to COMPLETED / CANCELLED_BY_RIDER / CANCELLED_BY_DRIVER
trip_fare_breakdown	trip_id	trip_id to trip_headers	Amounts, distance, duration and surge >= 0; payment_method limited to five values
trip_reviews	review_id	trip_id to trip_headers	Both ratings between 1 and 5

Rand amounts use NUMERIC (10,2) so they are exact, identifiers are VARCHAR, trip times are TIMESTAMP, and 22 CHECK constraints keep categories and amounts inside valid domains. NOT NULL is applied to the fields every research question depends on.

Relationships

Parent table	Child table	Link	Cardinality
sa_drivers	vehicles	driver_id	One-to-many (the data holds exactly one vehicle per driver)
sa_drivers	trip_headers	driver_id	One-to-many
sa_riders	trip_headers	rider_id	One to many
pricing_surge_zones	trip_headers	zone_id	One to many
trip_headers	trip_fare_breakdown	trip_id	One-to-one (shared primary key)
trip_headers	trip_reviews	trip_id	One-to-many (about half of trips have a review)

Tools used

Database and analysis: PostgreSQL with pgAdmin 4 (tables, ERD, query results, and the Graph Visualiser charts for queries). Presentation: Microsoft PowerPoint, using native charts. Report figures: Python (matplotlib) and Graphviz. Version control and file sharing: GitHub.

3. SQL development and performance

Technique	RQ1	RQ2	RQ3	RQ4	RQ5	RQ6
Multi-table joins	●	●	●	●	●	●
LEFT JOIN			●			
GROUP BY and aggregates	●	●	●	●	●	●
HAVING			●		●	
CTEs		●	●	●	●	●
Subqueries			●			●
Window functions (RANK, DENSE_RANK, AVG OVER)		●	●		●	
PERCENTILE_CONT and CORR			●	●		
CASE and EXTRACT				●		●
READ-ONLY transactions and TEMP tables			●			●
NULLIF guards on every division	●	●	●	●	●	●

The matrix reflects the techniques in the saved SQL files. Plus additional_advanced_queries.sql: each platform's share of completed trips (Uber 35.00%, Dual-Platform 32.95%, Bolt 32.05%), city ranks, a monthly trend using LAG (average payout stays between R356.74 in August 2025 and R371.75 in February 2026), and a subquery that lists drivers above the overall average.

Design decisions in the SQL

- Completed trips only: cancelled trips have no fare, so every earnings query filters for trip_status = 'COMPLETED'.
- RQ3 reports both the average of per-trip ratios and total payout divided by total fare, so the weighting is explicit.
- RQ6 divides total earnings by total trip hours; RQ4 and RQ5 average the per-trip hourly figure. The two methods differ slightly and are not mixed.
- RQ5 keeps only city and category combinations with at least 30 completed trips.
- RQ6.sql uses TEMP tables, so the whole file runs in one database session; one fuel-price table feeds every RQ6 comparison.

Selected SQL

RQ3: categories above the overall average payout (HAVING with a scalar subquery)

```
HAVING AVG(driver_payout_zar) > (
    SELECT AVG(driver_payout_zar)
    FROM eligible_trips
)
```

RQ2: ranking and difference from the platform average (window functions)

```
RANK() OVER (
    ORDER BY avg_retention_pct_of_commissionable_fare DESC
) AS retention_rank,
ROUND(avg_retention_pct_of_commissionable_fare
      - AVG(avg_retention_pct_of_commissionable_fare) OVER (), 2)
      AS difference_from_platform_average_pp
```

RQ6: estimated fuel cost and net earnings per trip

```
distance_km * litres_per_100km / 100 * fuel_price_per_litre
    AS estimated_fuel_cost,
driver_payout_zar - (distance_km * litres_per_100km / 100
    * fuel_price_per_litre) AS estimated_net_earnings
```

The complete queries are in 02_SQL_Queries (RQ1.sql to RQ6.sql and additional_advanced_queries.sql).

Performance testing (RQ4 queries)

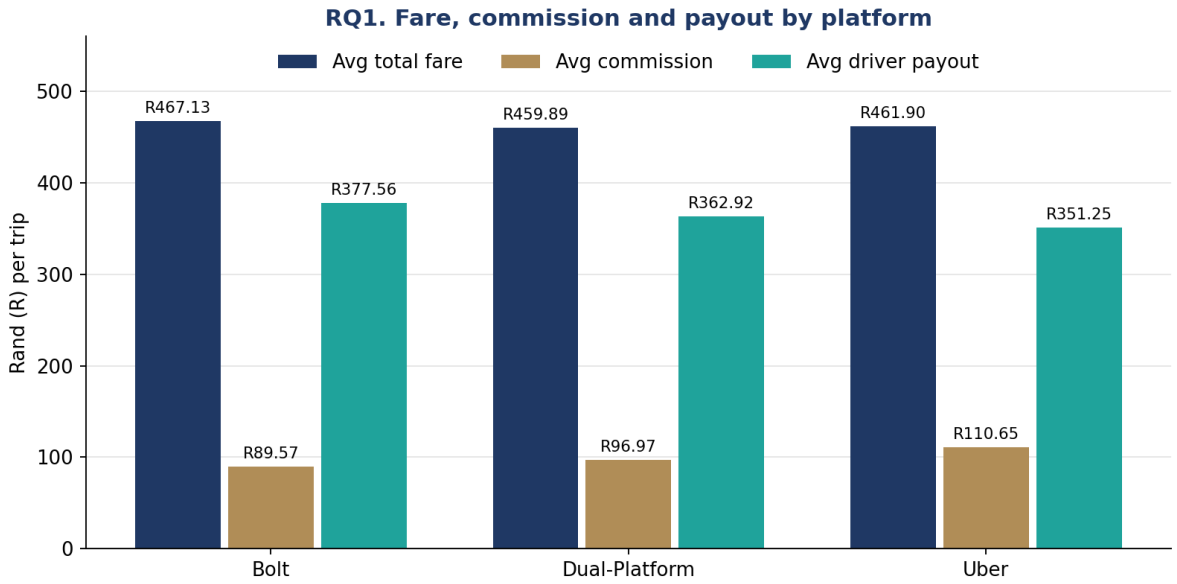
Measure	4A: peak vs off-peak	4E: peak x surge (CTE)
Execution time	117.0 ms	230.0 ms
Planning time	0.41 ms	0.44 ms
Rows processed	33,331	33,331
Table scans	Sequential on both tables	Sequential on both tables
Join method	Hash join, in memory	Hash join, in memory
Disk reads	None	None

PostgreSQL chose sequential scans over the trip_status index because the COMPLETED filter matches about 67% of the table (33,331 of 50,000 rows). Reading the whole table is cheaper than index lookups at that selectivity, so the planner's choice was correct. The join, sort, and aggregation all ran in memory. At 0.1 to 0.2 seconds, no further optimisation was needed.

4. Findings by research question

RQ1: Fares, commission and payout by platform

Method: sa_drivers, trip_headers and trip_fare_breakdown are joined and filtered to completed trips; averages are grouped by platform affiliation.



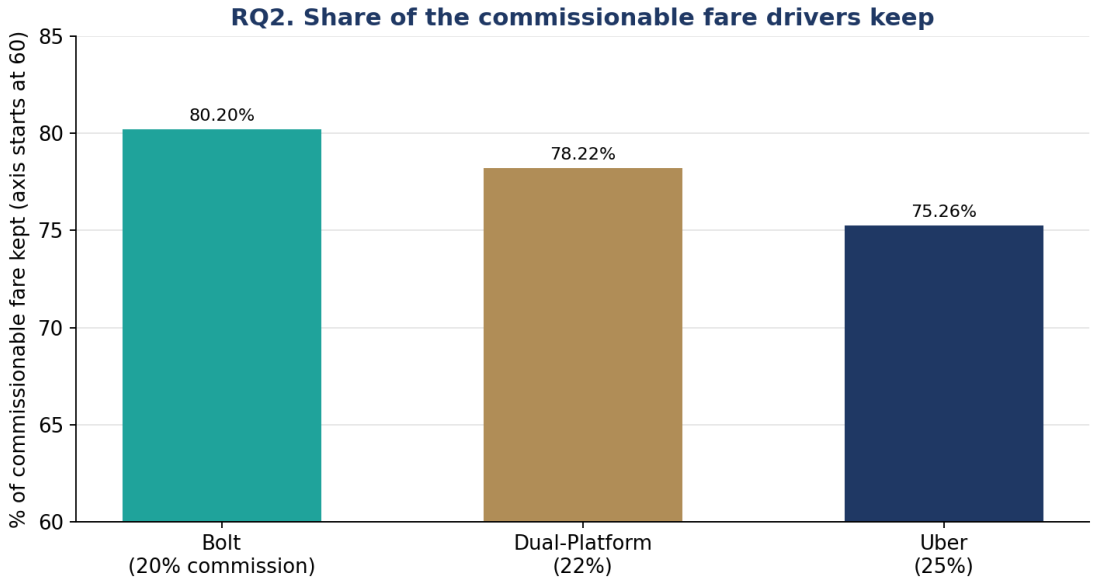
Source: completed trips, cmpg321_data_alchemists database

Platform	Completed trips	Avg total fare	Avg commission	Avg driver payout
Bolt	10,681	R467.13	R89.57	R377.56
Dual-Platform	10,983	R459.89	R96.97	R362.92
Uber	11,667	R461.90	R110.65	R351.25

Bolt drivers received R26.31 more per trip than Uber drivers. Only R5.23 of that comes from a higher average fare; the other R21.08 comes from a lower commission. Dual-Platform drivers sit between the two on commission and payout and have the lowest average fare. These are associations in the dataset, not proof that the platform alone causes the difference.

RQ2: Commission and the share of the fare drivers keep

Method: retention = (payout minus tip) divided by (total fare minus tip), averaged per trip and grouped by commission rate; platforms are ranked with RANK () and compared with the platform average using a window function.



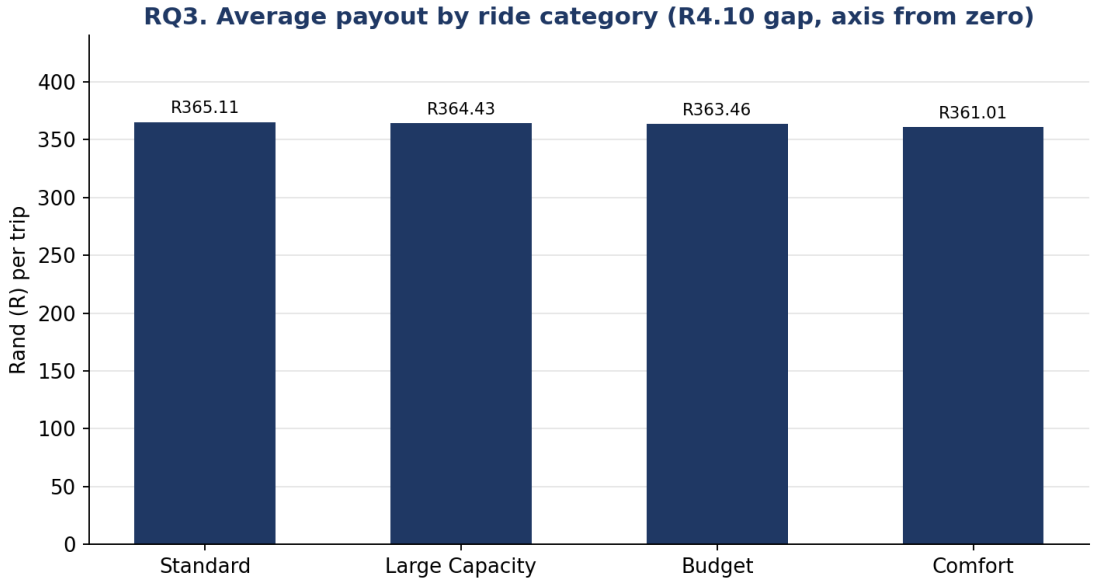
Source: completed trips, cmpg321_data_alchemists database

Platform	Commission rate	Share of commissionable fare kept
Bolt	20%	80.20%
Dual-Platform	22%	78.22%
Uber	25%	75.26%

A higher commission rate goes with a lower retained share, and the 4.94-point gap between Bolt and Uber tracks the 5-point commission gap. Retention is (payout minus tip) divided by (total fare minus tip), averaged per trip, because commission is calculated on the fare without tips and tips reach the driver in full.

RQ3: Ride category

Method: completed trips with a positive fare are grouped by ride category; both the average of per-trip ratios and total payout divided by total fare are reported, with medians from PERCENTILE_CONT.



Source: completed trips, cmpg321_data_alchemists database

Category	Completed trips	Avg driver payout	Avg payout-to-fare ratio
Standard	8,507	R365.11	78.92%

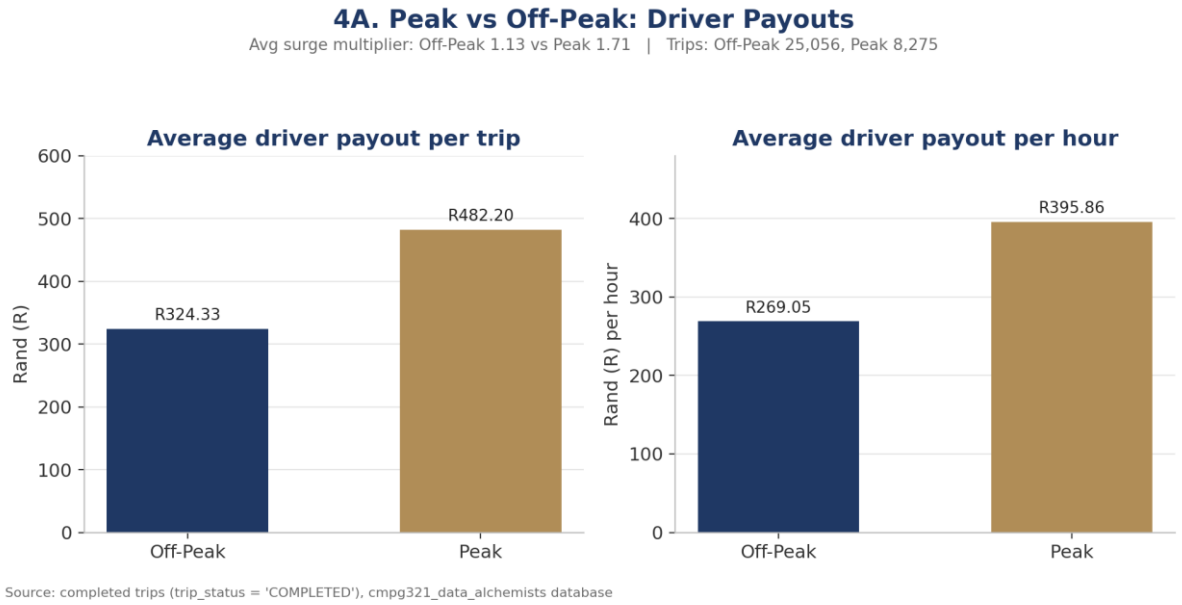
Large Capacity	8,334	R364.43	78.92%
Budget	8,386	R363.46	78.89%
Comfort	8,104	R361.01	78.87%

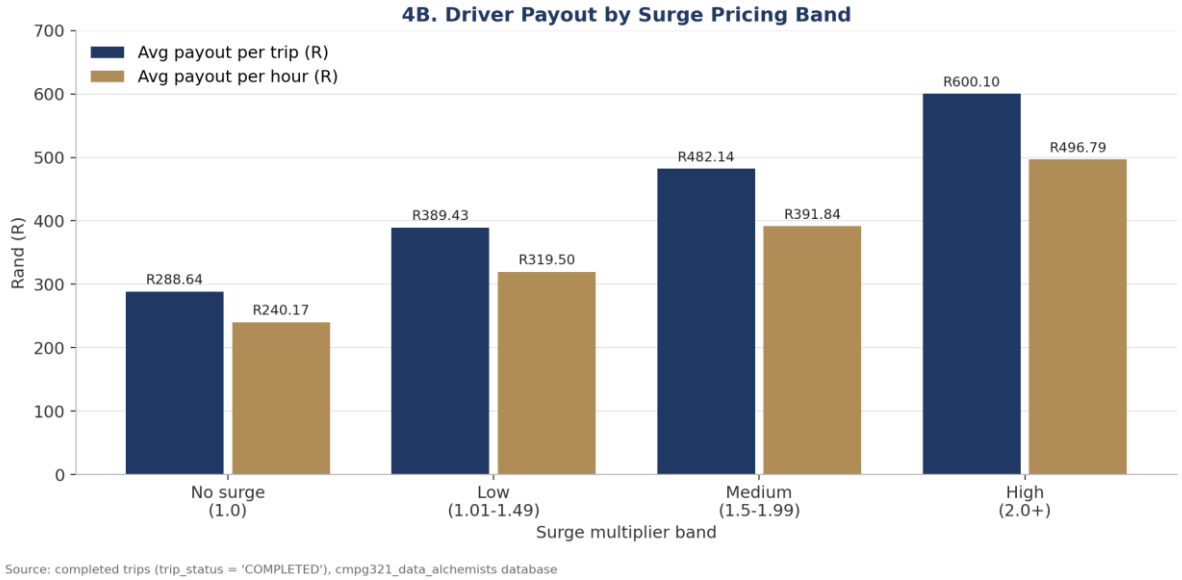
Standard pays the most, but only R4.10 separates the best and worst categories, and drivers keep about 79% of the fare in every category. Ride category has very little effect on what a driver earns.

RQ4: Distance, duration, peak periods and surge pricing

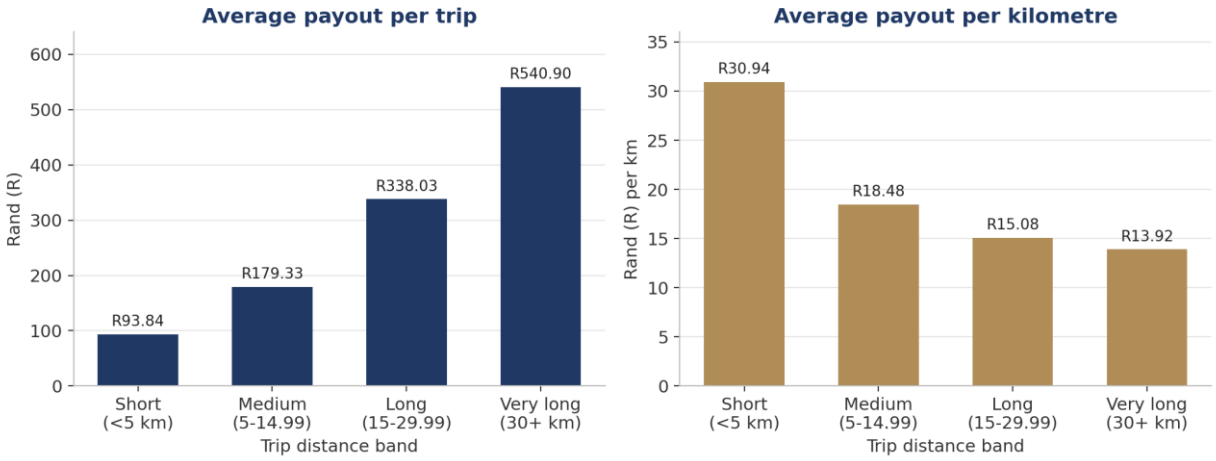
Method: completed trips are banded by surge, distance, and duration with CASE; peak is 06:00-08:59 and 16:00-18:59, found with EXTRACT; CORR measures each factor's correlation with payout.

Distance (0.754), duration (0.735), and surge (0.527) correlate with payout; tips barely do (0.074). Longer trips pay more per trip but less per kilometre and per hour. Peak trips pay R482.20 against R324.33 off-peak, but the cause is surge: 72% of peak trips fall in the medium or high surge bands, against 14% of off-peak trips. Within the same surge band, peak and off-peak payouts are almost identical.



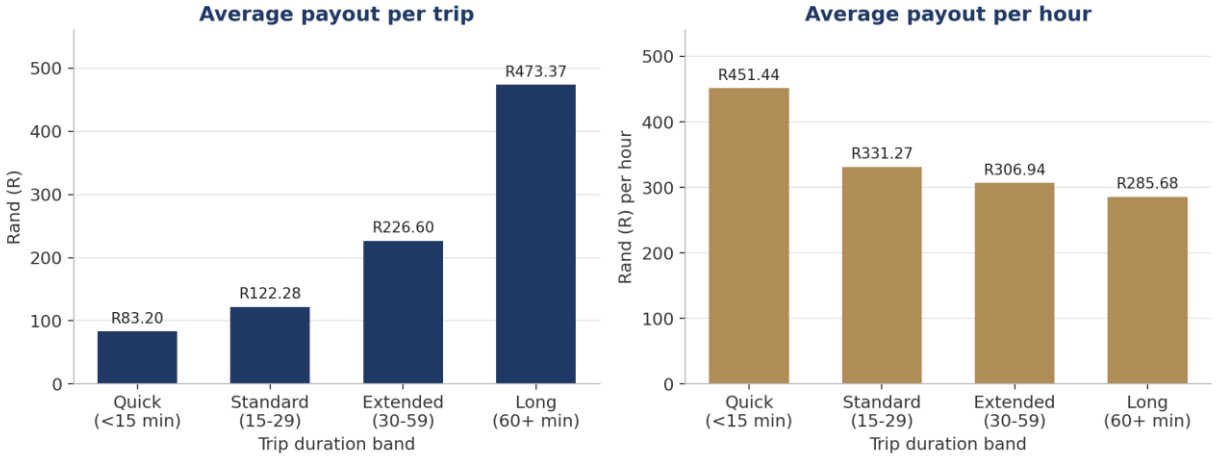


4C. Trip Distance: Higher Total Payout, Lower Payout per Km



Source: completed trips (trip_status = 'COMPLETED'), cmpg321_data_alchemists database

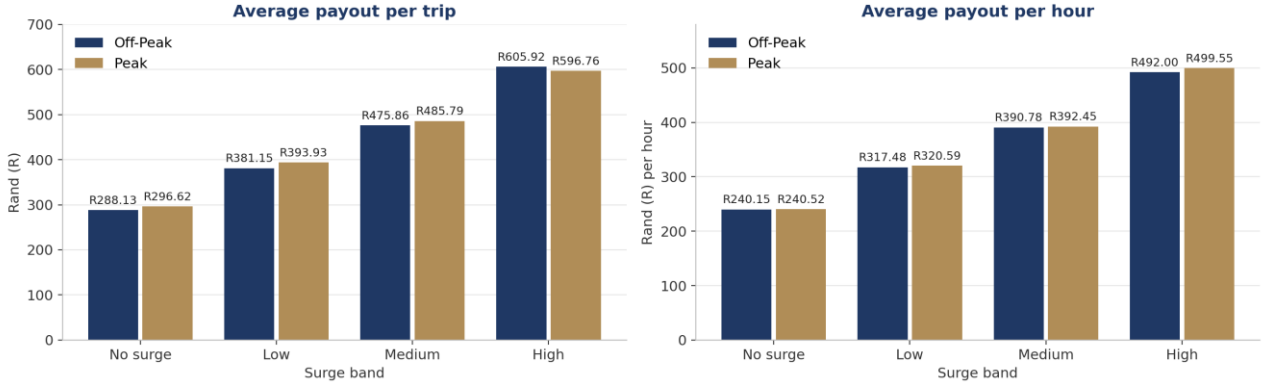
4D. Trip Duration: Higher Total Payout, Lower Hourly Rate



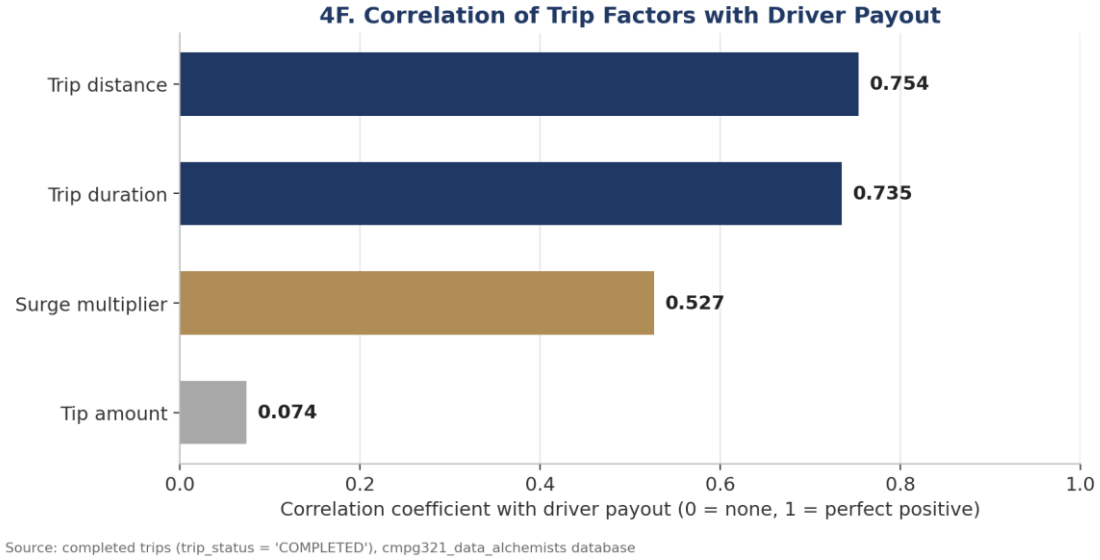
Source: completed trips (trip_status = 'COMPLETED'), cmpg321_data_alchemists database

4E. Peak vs Off-Peak Within Each Surge Band

Within the same surge band, peak and off-peak payouts are almost identical



Source: completed trips (trip_status = 'COMPLETED'), cmpg321_data_alchemists database



RQ5: City and ride-category combination

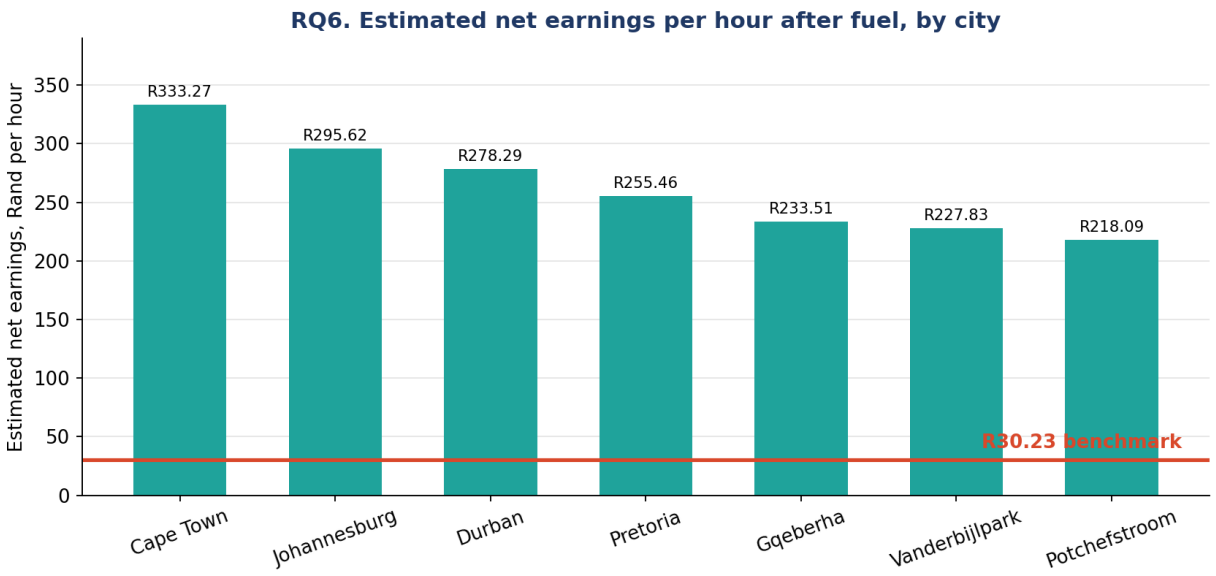
Method: the city comes from the trip's pricing zone, not the driver's home city; combinations with fewer than 30 completed trips are excluded, and the rest are ranked with RANK() and DENSE_RANK().

City	Budget	Standard	Comfort	Large Capacity
Cape Town	R456.41	R443.77	R442.38	R449.28
Johannesburg	R400.02	R419.39	R402.89	R409.34
Durban	R385.57	R393.76	R380.59	R376.78
Pretoria	R354.55	R352.96	R350.75	R368.87
Gqeberha	R319.51	R327.66	R332.02	R321.12
Vanderbijlpark	R322.09	R319.24	R324.75	R327.45
Potchefstroom	R310.97	R300.31	R306.55	R302.76

Average payout per completed trip, by city and ride category. Cape Town takes the top four places, with Large Capacity best per kilometre (R20.71) and per hour (R375.86). Cape Town occupies the top four places. The R22.99 gap between Cape Town's weakest and Johannesburg's strongest combination is larger than the R14.03 spread between the best and worst category within Cape Town, so the city matters more than the ride category. The best category also depends on the measure: raw payout, payout per kilometre, and payout per hour do not give the same ranking.

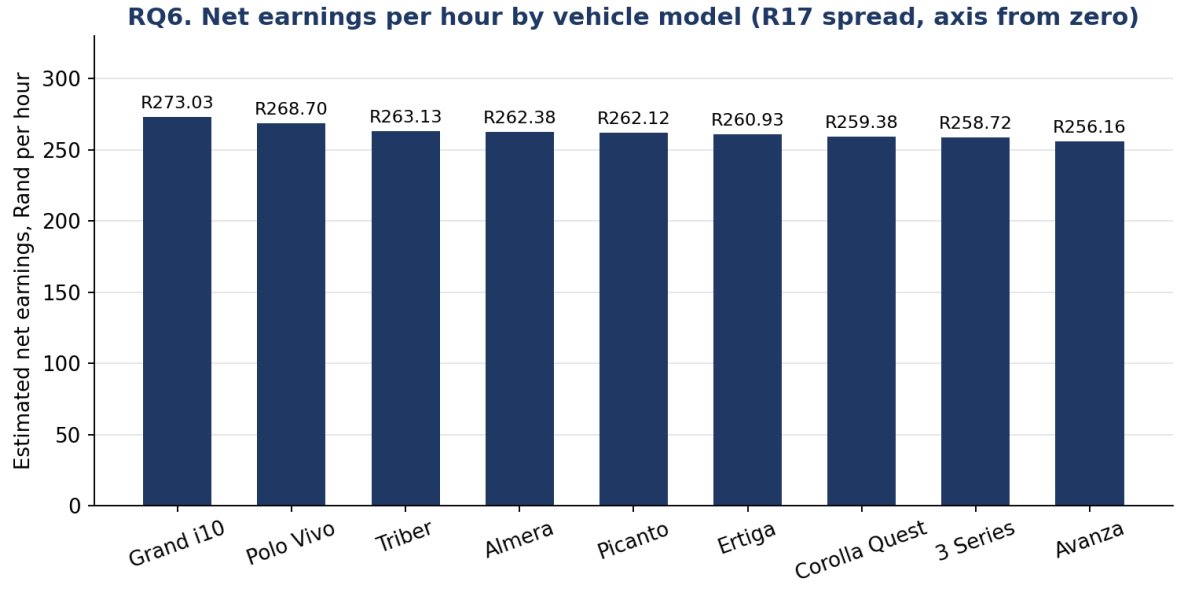
RQ6: Net hourly earnings after estimated fuel

Method: fuel cost = distance x litres per 100 km / 100 x fuel price, using each vehicle model's consumption and the monthly Petrol 95 price for the trip's region (inland: Johannesburg, Pretoria, Vanderbijlpark, Potchefstroom; coastal: Cape Town, Durban, Gqeberha); net earnings per hour = total net earnings divided by total trip hours.



Source: completed trips, cmpg321_data_alchemists database

City	Avg net earnings per trip	Est. net earnings per hour
Cape Town	R417.77	R333.27
Johannesburg	R376.41	R295.62
Durban	R353.23	R278.29
Pretoria	R325.04	R255.46
Gqeberha	R294.68	R233.51
Vanderbijlpark	R291.10	R227.83
Potchefstroom	R273.58	R218.09



Source: completed trips, cmpg321_data_alchemists database

Across the nine vehicle models, estimated net earnings only range from R256.16 (Avanza) to R273.03 (Grand i10), a spread of about R17, against a spread of about R115 across cities. Where a driver operates matters far more than the car they drive.

Overall estimated net earnings are R262.52 per hour (R287.22 before fuel), far above the benchmarks of R28.79 per hour (March 2025 to February 2026) and R30.23 (from March 2026), and all seven cities clear

them. Fuel is a small cost (about R31 per trip); differences between cities come mainly from payouts and trip duration. These are estimates after fuel only, based on recorded trip time. They exclude waiting time, maintenance, insurance, depreciation and financing, so they must not be read as profit or as proof of minimum-wage compliance.

5. Actionable insights

- Drivers: commission is the clearest lever (Bolt keeps about R26 more per trip than Uber). Operate in Cape Town where possible. Short trips pay more per kilometre and per hour; long and high-surge trips pay more per trip.
- Platforms: a 5-point commission gap moves driver retention by about 5 points. Ride category is not a differentiator. Surge is where payout rises fastest.
- Policy and advocacy: headline hourly pay clears R30.23 after fuel, but idle time and vehicle costs are unmeasured. Utilisation data is needed before calling this a living wage.

6. Limitations

- Fuel cost uses model-level consumption assumptions and monthly Petrol 95 prices, so net earnings are estimates, not profit.
- Hourly figures count trip time only, not waiting or idle time.
- Maintenance, insurance, depreciation, tyres and financing are not deducted.
- Peak hours (06:00-08:59 and 16:00-18:59) are a project-defined window, not a dataset field.
- Results show associations in the supplied data, not causation.

7. Conclusion

Drivers keep most of the fare, and after fuel, the average trip leaves them R332.26. Commission and city move their pay most; ride category barely does; surge, not time of day, explains the peak premium. The honest answer to our opening question is that the numbers look healthy on paper, but only utilisation and vehicle-cost data can say whether they add up to a living wage.

Appendix A: Data dictionary

Field types and ranges as profiled from the supplied files; keys and constraints are summarised in the table design in Section 2.

sa_drivers (800 rows)

Field	Type	Meaning and domain
driver_id (PK)	Text	e.g. DRV_00001
first_name, last_name	Text	Driver name
operating_city	Text	One of seven South African cities
platform_affiliation	Text	Uber, Bolt or Dual-Platform
platform_commission_pct	Number	Commission rate: 20%, 22% or 25%
driver_rating	Number	About 4.10 to 5.00
total_lifetime_trips	Integer	About 181 to 6,795
onboarding_date	Date	YYYY/MM/DD

vehicles (800 rows)

Field	Type	Meaning and domain
vehicle_id (PK)	Text	e.g. VEH_00001
driver_id (FK)	Text	Owning driver
make, model	Text	Nine models across eight makes
year	Integer	2016 to 2025
license_plate (UNIQUE)	Text	South African registration format
category_type	Text	Standard, Comfort or Large Capacity

sa_riders (1,200 rows)

Field	Type	Meaning and domain
rider_id (PK)	Text	Unique rider identifier
first_name, last_name, mobile_number	Text	Rider profile
home_city, province	Text	Rider location
preferred_payment	Text	Preferred payment method
account_created_date	Date	Account creation date

pricing_surge_zones (9 rows)

Field	Type	Meaning and domain
zone_id (PK)	Text	Unique zone identifier
zone_name, city	Text	Zone description and city
base_fare_zar, per_km_rate_zar, per_min_rate_zar	Number	Zone tariff in rand

trip_headers (50,000 rows)

Field	Type	Meaning and domain
trip_id (PK)	Text	e.g. TRIP_2026_0039559
rider_id, driver_id, zone_id (FK)	Text	Links to rider, driver and zone
ride_category	Text	Budget, Standard, Comfort or Large Capacity
request_timestamp	Timestamp	YYYY-MM-DD HH:MM:SS, 1 March 2025 to 31 March 2026
trip_status	Text	COMPLETED (33,331), CANCELLED_BY_RIDER (8,417) or CANCELLED_BY_DRIVER (8,252)
cancellation_reason	Text	Seven reasons; empty for completed trips

trip_fare_breakdown (50,000 rows)

Field	Type	Meaning and domain
trip_id (PK, FK)	Text	One row per trip
distance_km	Number	0 to 48 km
duration_minutes	Integer	0 to 182 minutes
surge_multiplier	Number	1.0 to 2.4
base_fare_zar, tolls_zar, tip_zar	Number	Fare components in rand
total_fare_zar	Number	Up to R1,988.22
platform_commission_zar	Number	Up to R494.59; calculated on the fare without tips
driver_payout_zar	Number	Up to R1,550.11; total fare minus commission
payment_method	Text	Cash, debit card, credit card, EFT (Ozow) or in-app wallet

trip_reviews (25,055 rows)

Field	Type	Meaning and domain
review_id (PK)	Text	Unique review identifier
trip_id (FK)	Text	Reviewed trip
rider_rating_of_driver	Integer	1 to 5
driver_rating_of_rider	Integer	3 to 5
feedback_comment	Text	Short feedback text